

CSC3003S Capstone Project — Stage One

Risk [30 Marks]

Project Abbrev	BIOFILM2	Client/Supervisor + email	Michelle Kuttel mkuttel@cs.uct.ac.za
Date	08/08/2025	Tutor + email	Aidan Ravenscroft RVNAID001@myuct.ac.za
Team members	Byron Clarke CLRBYP003	Ruth-ann Rudolph RDLRUT001	Sonali Naidoo NDXSON001

Risk description [1]	Consequence [1]	Prob [1/2]	Impact [1/2]	Strategy (ARTA) [1/2]	Metric [½]+ monitoring [½]	Management [1½]
Overcomplicating mathematical modelling for the simulation and being forced to redesign.	Need to redesign many formulas and redo code to work with new calculations	low	high	Avoid	Review formulas with team members to confirm confidence in their simplicity	Mathematics for the model are discussed and agreed upon to be sufficiently simple to avoid unnecessary complexity
Bad simulation performance	Simulations take much longer to run than expected	low	high	Transfer	Track time to run simulation at different stages in development: Runtime to get to a specific number of bacteria	Speak to client and require they accept this as a possibility
Difference in understanding of model features and flexibility with the client	We waste time creating unwanted features that may need to be removed at a later stage	high	low	Reduce	Get list of expectations from client and compare all model features against client expectations	Speak to client early on, to compare and then align expectations for features
Team scheduling conflicts	Productivity delayed since teammates rely on work done by others	high	low	Reduce	Track frequency of rescheduled meetings and tasks: Rescheduled items per week	Plan team meetings, and requirements in advance to allow for rescheduling if needed. Do majority of work asynchronously
Scope is too large to implement before the deadline	Code incomplete and unusable at deadline	low	high	Avoid	Track the rate that features are completed: features per week	The scope is reduced appropriately

Team member is unable to contribute for a time	Productivity on the project decreases greatly	low	high	Reduce	Monitor team members well-being. Be wary for if any start to feel sick or may face personal emergencies.	Deadlines must be delegated with leeway, which can be leveraged in unexpected productivity drops
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